

OCR & Text Extraction Workflow

| Tool | Purpose | Notes |

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| **Tesseract** | OCR engine for handwritten/printed text | Open source, supports multiple languages |

| **Google Vision API** | Cloud based OCR & image analysis | Reduces cost & improves accuracy for large volumes |

| **Codex** | Code generation & debugging assistant | Helps rewrite buggy snippets (e.g., Two Sum) |

| **Gemini** | Google's generative AI (Chat style) | Useful for quick explanations & code reviews |

> **Next big step** – Once OCR extraction is stable, store the raw text in a database for downstream processing (search, analytics, etc.).

Project Management & Collaboration

- **Git Ignore File Basics** – Keep build artefacts out of version control.

- **Pinned Items** – Highlight critical tasks (e.g., interview prep, trading platform).

- **Chat & Communication**

- **ChatGPT** (via `chatgpt.com`)

- **Gemini** (via `gemini.google.com`)

- Slack channels: `Clusters`, `Atlas`, `Pranathi Stikr`

- **Task Tracking**

- `Projects` → `Projects aif confidence low` (likely a typo for **AI** confidence)

- `Library` → `38 Apps Google Vision API`

- `Upgrade` → `Chat@PT v + Upgrade 7, Share`

Algorithms & Data Structures

| Topic | Key Points |

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| **Two Sum** | Debugging strategy: use hash map for $O(n)$ time. |

| **Sliding Window – First Negative** | Find the first negative number in every window of size k . |

| **Linked List Binary Conversion** | Convert a binary number represented as a linked list into an integer. |

| **Tesseract Integration** | Wrap OCR calls in a reusable module; handle errors & retries. |

Development Practices

- **Full Stack Cover Letter** – Template for job applications.

- **Best Coding Practices** – Clean code, meaningful variable names, unit tests.

- **Git Workflow** – Commit often, use feature branches, pull requests.

- **Database Design** – Store OCR results with metadata (timestamp, source, confidence).

Miscellaneous Topics

| Item | Context |

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| **Year 1895 Calendar** | Historical reference; could be used for data-driven projects. |

| **Club Application & Gaming in India** | Potential project idea or case study. |

| **Employment Verification Follow-up** | Process for verifying candidate credentials. |

| **Ask Gemini** | Quick AI query for clarifications. |

| **Chat@PT v + Upgrade 7** | Likely a note to upgrade the ChatGPT API version. |

Action Items

1. **Finalize OCR pipeline** – Tesseract → Google Vision → Codex for post-processing.
2. **Persist OCR output** – Store in a relational or NoSQL database with indexing.
3. **Set up Git ignore** – Add build artefacts, logs, and OCR temp files.
4. **Implement algorithm prototypes** – Two-Sum, Sliding Window, Linked List conversion.
5. **Prepare interview material** – Focus on OCR, AI integration, and algorithmic challenges.
6. **Plan trading platform** – Define data sources, real-time processing, and UI.

> **Tip:** Keep your notes in Markdown for easy conversion to PDFs or slides. Use headings (`#`, `##`, `###`) to structure topics, bullet points for clarity, and code fences for snippets.